Energy Methods and Their Application

Uniqueness of Initial Boundary Value Problems

Souvik Ghorui
Roll No. M25MA2008

Department of Mathematics
Course: MAL5030 (M.Sc. M.Tech)

March 29, 2026

Motivation

- Physical phenomena (heat, waves) are modelled by PDEs.

Motivation

- Physical phenomena (heat, waves) are modelled by PDEs.
- We need to establish **well-posedness**.

Motivation

- Physical phenomena (heat, waves) are modelled by PDEs.
- We need to establish **well-posedness**.
- Classical techniques rely on explicit solutions.

Motivation

- Physical phenomena (heat, waves) are modelled by PDEs.
- We need to establish **well-posedness**.
- Classical techniques rely on explicit solutions.
- **Goal:** A unified, flexible approach based directly on the PDE structure.

Hadamard's Well-Posedness

A PDE problem is well-posed if a solution:

- **Exists**

Hadamard's Well-Posedness

A PDE problem is well-posed if a solution:

- **Exists**
- **Is Unique**

Hadamard's Well-Posedness

A PDE problem is well-posed if a solution:

- **Exists**
- **Is Unique**
- **Depends continuously on data** (Stability)

Hadamard's Well-Posedness

A PDE problem is well-posed if a solution:

- **Exists**
- **Is Unique**
- **Depends continuously on data** (Stability)

Hadamard's Well-Posedness

A PDE problem is well-posed if a solution:

- **Exists**
- **Is Unique**
- **Depends continuously on data** (Stability)

The Energy Method provides elegant proofs for Uniqueness and Stability.

Mathematical Background

- L^2 **Norm:** Measures global size.

$$\|f\|^2 = \int_0^L [f(x)]^2 dx$$

Mathematical Background

- L^2 **Norm:** Measures global size.

$$\|f\|^2 = \int_0^L [f(x)]^2 dx$$

- **Integration by Parts (IBP):** Key for spatial derivatives.

$$\int_0^L u_{xx} u \, dx = [u_x u]_0^L - \int_0^L u_x^2 dx$$

Mathematical Background

- L^2 **Norm:** Measures global size.

$$\|f\|^2 = \int_0^L [f(x)]^2 dx$$

- **Integration by Parts (IBP):** Key for spatial derivatives.

$$\int_0^L u_{xx} u \, dx = [u_x u]_0^L - \int_0^L u_x^2 dx$$

- **Gronwall's Inequality:** Bounds energy growth over time.

$$E'(t) \leq CE(t) \implies E(t) \leq E(0)e^{Ct}$$

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**
  - Multiply the PDE by u (or u_t).

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**
  - Multiply the PDE by u (or u_t).
  - Integrate over the domain.

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**
  - Multiply the PDE by u (or u_t).
  - Integrate over the domain.
  - Apply IBP and boundary conditions.

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**
  - Multiply the PDE by u (or u_t).
  - Integrate over the domain.
  - Apply IBP and boundary conditions.
- **Step 2: Apply Gronwall**

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**
  - Multiply the PDE by u (or u_t).
  - Integrate over the domain.
  - Apply IBP and boundary conditions.
- **Step 2: Apply Gronwall**
  - Formulate an inequality: $E'(t) \leq CE(t) + G(t)$.

The Energy Method: Two-Step Strategy

- **Step 1: Energy Identity**
  - Multiply the PDE by u (or u_t).
  - Integrate over the domain.
  - Apply IBP and boundary conditions.
- **Step 2: Apply Gronwall**
  - Formulate an inequality: $E'(t) \leq CE(t) + G(t)$.
  - Deduce the explicit bound for $E(t)$.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.
- Define the difference: $w = u_1 - u_2$.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.
- Define the difference: $w = u_1 - u_2$.
- w satisfies a **homogeneous** IBVP with $w(x, 0) = 0$.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.
- Define the difference: $w = u_1 - u_2$.
- w satisfies a **homogeneous** IBVP with $w(x, 0) = 0$.
- Define energy $E_w(t) \geq 0$.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.
- Define the difference: $w = u_1 - u_2$.
- w satisfies a **homogeneous** IBVP with $w(x, 0) = 0$.
- Define energy $E_w(t) \geq 0$.
- Show $E_w'(t) \leq 0$.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.
- Define the difference: $w = u_1 - u_2$.
- w satisfies a **homogeneous** IBVP with $w(x, 0) = 0$.
- Define energy $E_w(t) \geq 0$.
- Show $E_w'(t) \leq 0$.
- Since $E_w(0) = 0$, it follows that $E_w(t) = 0$ for all $t \geq 0$.

How Uniqueness Follows

- Assume two solutions: u_1 and u_2.
- Define the difference: $w = u_1 - u_2$.
- w satisfies a **homogeneous** IBVP with $w(x, 0) = 0$.
- Define energy $E_w(t) \geq 0$.
- Show $E_w'(t) \leq 0$.
- Since $E_w(0) = 0$, it follows that $E_w(t) = 0$ for all $t \geq 0$.
- Conclusion: $w \equiv 0 \implies u_1 = u_2$.

Application 1: Heat Equation

- **PDE:** $u_t = ku_{xx} + f(x, t)$

Application 1: Heat Equation

- **PDE:** $u_t = ku_{xx} + f(x,t)$
- **Energy Functional:**

$$E(t) = \frac{1}{2}\|u(\cdot,t)\|^2$$

Application 1: Heat Equation

- **PDE:** $u_t = ku_{xx} + f(x,t)$
- **Energy Functional:**

$$E(t) = \frac{1}{2}\|u(\cdot,t)\|^2$$

- **Energy Identity:**

$$E'(t) = -k\|u_x\|^2 + \langle u, f\rangle$$

Application 1: Heat Equation

- **PDE:** $u_t = ku_{xx} + f(x,t)$
- **Energy Functional:**

$$E(t) = \frac{1}{2}\|u(\cdot,t)\|^2$$

- **Energy Identity:**

$$E'(t) = -k\|u_x\|^2 + \langle u, f\rangle$$

- **Homogeneous Case ($f = 0$):** $E'(t) \leq 0$. Energy dissipates.

Application 1: Heat Equation

- **PDE:** $u_t = ku_{xx} + f(x,t)$
- **Energy Functional:**

$$E(t) = \frac{1}{2}\|u(\cdot,t)\|^2$$

- **Energy Identity:**

$$E'(t) = -k\|u_x\|^2 + \langle u, f\rangle$$

- **Homogeneous Case ($f = 0$):** $E'(t) \leq 0$. Energy dissipates.
- **Decay:**

$$E(t) \leq E(0)e^{-2k\pi^2 t/L^2}$$

Application 2: Wave Equation

- **PDE:** $u_{tt} = c^2 u_{xx} + f(x, t)$

Application 2: Wave Equation

- **PDE:** $u_{tt} = c^2 u_{xx} + f(x,t)$
- **Energy Functional:** (Kinetic + Potential)

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

Application 2: Wave Equation

- **PDE:** $u_{tt} = c^2 u_{xx} + f(x,t)$
- **Energy Functional:** (Kinetic + Potential)

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

- **Energy Identity:**

$$E'(t) = \langle u_t, f \rangle$$

Application 2: Wave Equation

- **PDE:** $u_{tt} = c^2 u_{xx} + f(x,t)$
- **Energy Functional:** (Kinetic + Potential)

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

- **Energy Identity:**

$$E'(t) = \langle u_t, f \rangle$$

- **Homogeneous Case ($f = 0$):** $E'(t) = 0$.

Application 2: Wave Equation

- **PDE:** $u_{tt} = c^2 u_{xx} + f(x,t)$
- **Energy Functional:** (Kinetic + Potential)

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

- **Energy Identity:**

$$E'(t) = \langle u_t, f \rangle$$

- **Homogeneous Case ($f = 0$):** $E'(t) = 0$.
- **Result:** Energy is strictly conserved ($E(t) = E(0)$).

Example 1: Reaction-Diffusion Equation

- **PDE:** $u_t = u_{xx} + u \quad \text{for } x \in (0, \pi)$

Example 1: Reaction-Diffusion Equation

- **PDE:** $u_t = u_{xx} + u \quad$ for $x \in (0, \pi)$
- **Data:** $u(x, 0) = \sin x$, with zero Dirichlet boundary conditions.

Example 1: Reaction-Diffusion Equation

- **PDE:** $u_t = u_{xx} + u \quad$ for $x \in (0, \pi)$
- **Data:** $u(x, 0) = \sin x$, with zero Dirichlet boundary conditions.
- **Energy Identity:**

$$E'(t) = -\|u_x\|^2 + \|u\|^2$$

Example 1: Reaction-Diffusion Equation

- **PDE:** $u_t = u_{xx} + u \quad$ for $x \in (0, \pi)$
- **Data:** $u(x, 0) = \sin x$, with zero Dirichlet boundary conditions.
- **Energy Identity:**

$$E'(t) = -\|u_x\|^2 + \|u\|^2$$

- **Behavior:** Using the Poincaré inequality, $E'(t) \leq 0$. Energy is non-increasing.

Example 1: Reaction-Diffusion Equation

- **PDE:** $u_t = u_{xx} + u \quad$ for $x \in (0, \pi)$
- **Data:** $u(x, 0) = \sin x$, with zero Dirichlet boundary conditions.
- **Energy Identity:**

$$E'(t) = -\|u_x\|^2 + \|u\|^2$$

- **Behavior:** Using the Poincaré inequality, $E'(t) \leq 0$. Energy is non-increasing.
- **Exact Solution:** $u(x, t) = \sin x$
  (The energy remains constant at $\pi/4$).

Example 2: Damped Wave Equation

- **PDE:** $u_{tt} + 2\gamma u_t = c^2 u_{xx}$ $\quad$ ($\gamma > 0$ is the damping coefficient)

Example 2: Damped Wave Equation

- **PDE:** $u_{tt} + 2\gamma u_t = c^2 u_{xx}$ $\quad$ ($\gamma > 0$ is the damping coefficient)
- **Energy Functional:**

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

Example 2: Damped Wave Equation

- **PDE:** $u_{tt} + 2\gamma u_t = c^2 u_{xx}$ $\quad$ ($\gamma > 0$ is the damping coefficient)
- **Energy Functional:**

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

- **Energy Identity:**

$$E'(t) = -2\gamma\|u_t\|^2 \leq 0$$

Example 2: Damped Wave Equation

- **PDE:** $u_{tt} + 2\gamma u_t = c^2 u_{xx}$ $\quad$ ($\gamma > 0$ is the damping coefficient)
- **Energy Functional:**

$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

- **Energy Identity:**

$$E'(t) = -2\gamma\|u_t\|^2 \leq 0$$

- **Behavior:** The damping term strictly dissipates energy.

Example 2: Damped Wave Equation

- **PDE:** $u_{tt} + 2\gamma u_t = c^2 u_{xx}$ ($\gamma > 0$ is the damping coefficient)
- **Energy Functional:**

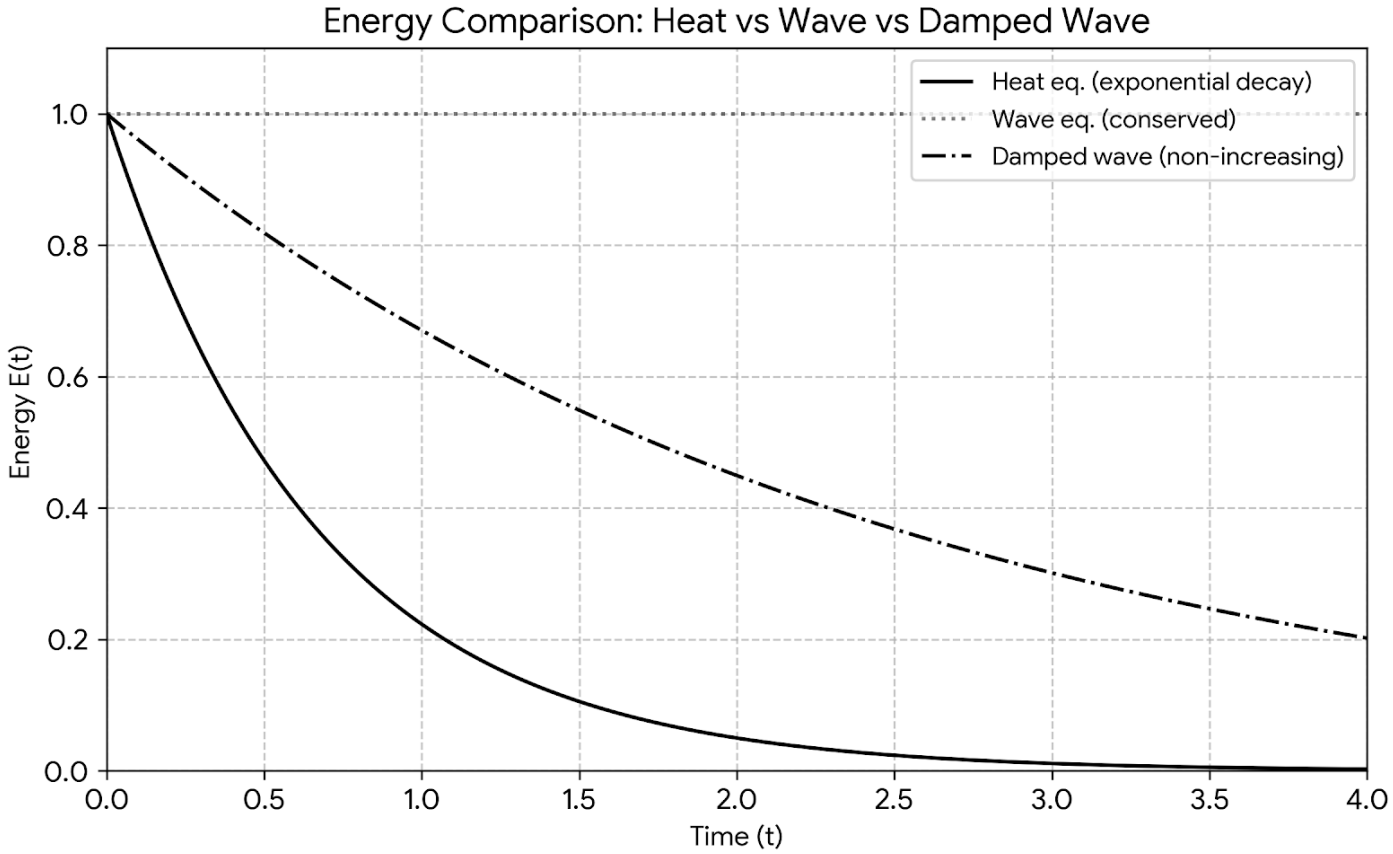
$$E(t) = \frac{1}{2}\int_0^L (u_t^2 + c^2 u_x^2)dx$$

- **Energy Identity:**

$$E'(t) = -2\gamma\|u_t\|^2 \leq 0$$

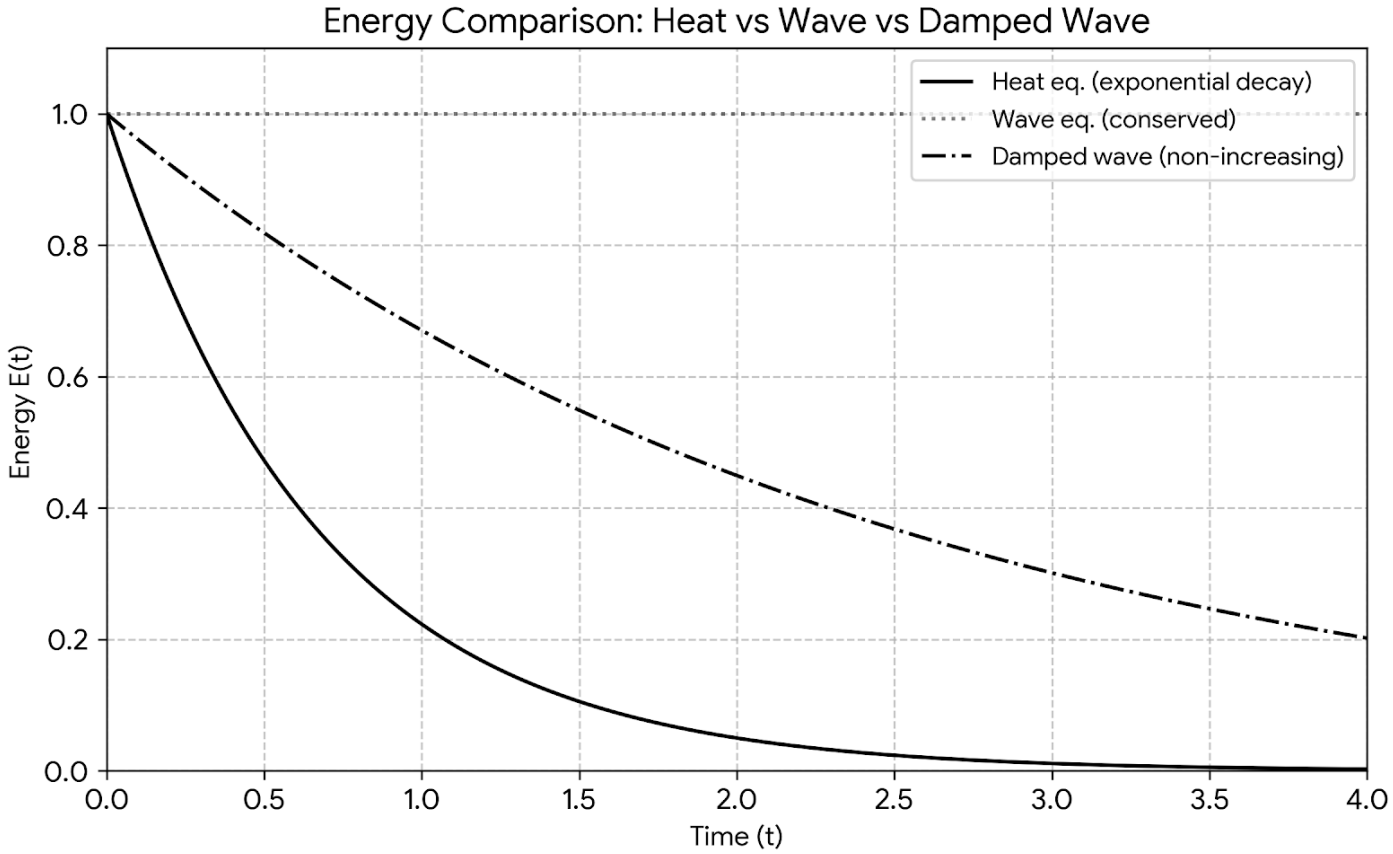
- **Behavior:** The damping term strictly dissipates energy.
- **Result:** The string gradually comes to rest, naturally forcing uniqueness.

Energy Behavior Over Time



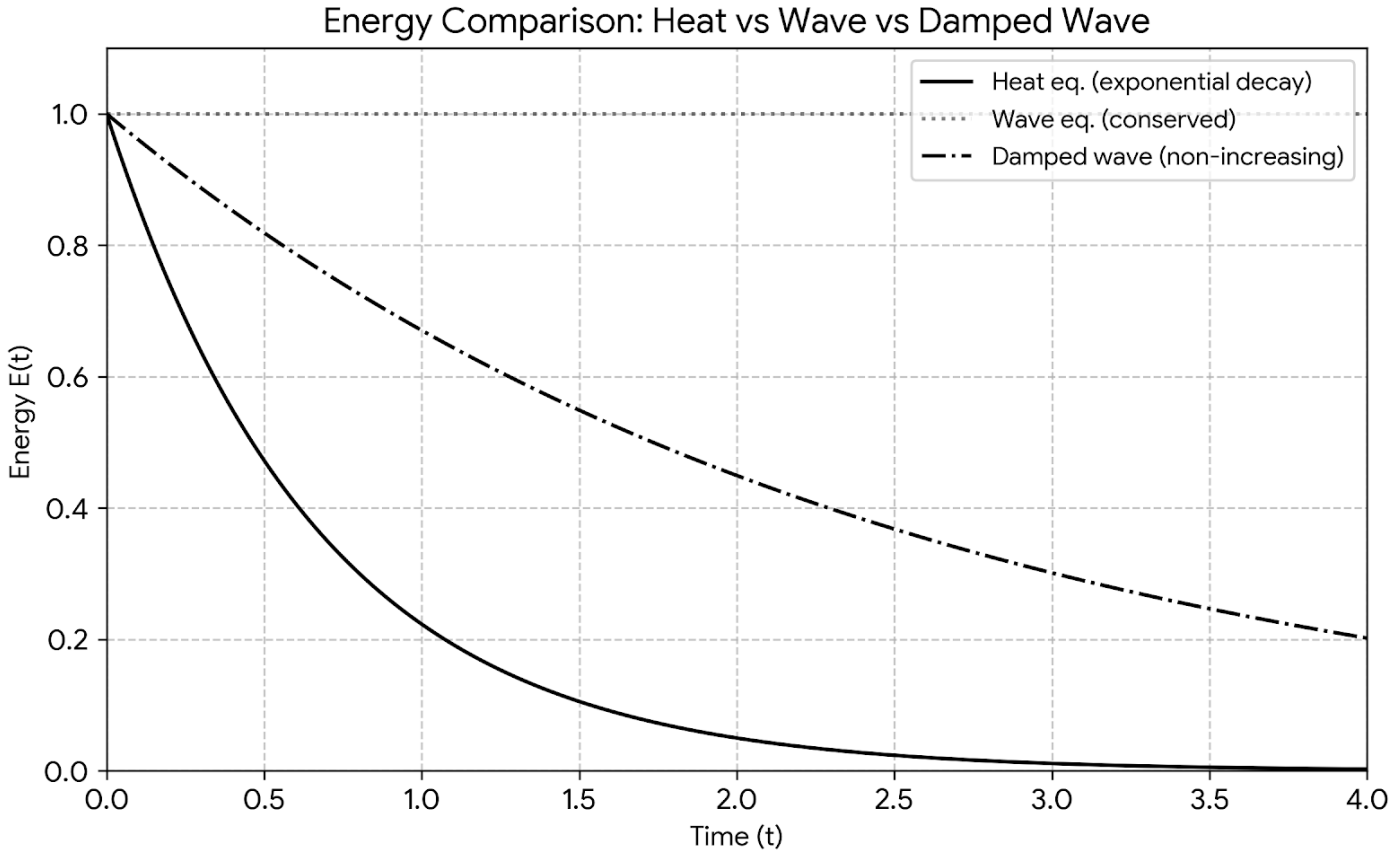
- **Heat Equation:** Energy dissipates exponentially.

Energy Behavior Over Time



- **Heat Equation:** Energy dissipates exponentially.
- **Wave Equation:** Energy is perfectly conserved.

Energy Behavior Over Time



- **Heat Equation:** Energy dissipates exponentially.
- **Wave Equation:** Energy is perfectly conserved.
- **Damped Wave Equation:** Damping continuously pulls energy out (non-increasing).

Discussion

Strengths:

- No explicit solution required.

Discussion

Strengths:

- No explicit solution required.
- Highly flexible (works for linear and many nonlinear PDEs).

Discussion

Strengths:

- No explicit solution required.
- Highly flexible (works for linear and many nonlinear PDEs).
- Provides direct physical insight.

Discussion

Strengths:

- No explicit solution required.
- Highly flexible (works for linear and many nonlinear PDEs).
- Provides direct physical insight.

Discussion

Strengths:

- No explicit solution required.
- Highly flexible (works for linear and many nonlinear PDEs).
- Provides direct physical insight.

Limitations:

- Primarily yields L^2 information, not pointwise maximums.

Discussion

Strengths:

- No explicit solution required.
- Highly flexible (works for linear and many nonlinear PDEs).
- Provides direct physical insight.

Limitations:

- Primarily yields L^2 information, not pointwise maximums.
- Deriving identities for complex nonlinear problems is difficult.

Discussion

Strengths:

- No explicit solution required.
- Highly flexible (works for linear and many nonlinear PDEs).
- Provides direct physical insight.

Limitations:

- Primarily yields L^2 information, not pointwise maximums.
- Deriving identities for complex nonlinear problems is difficult.
- Does not prove existence of a solution.

Conclusion

Thank You!